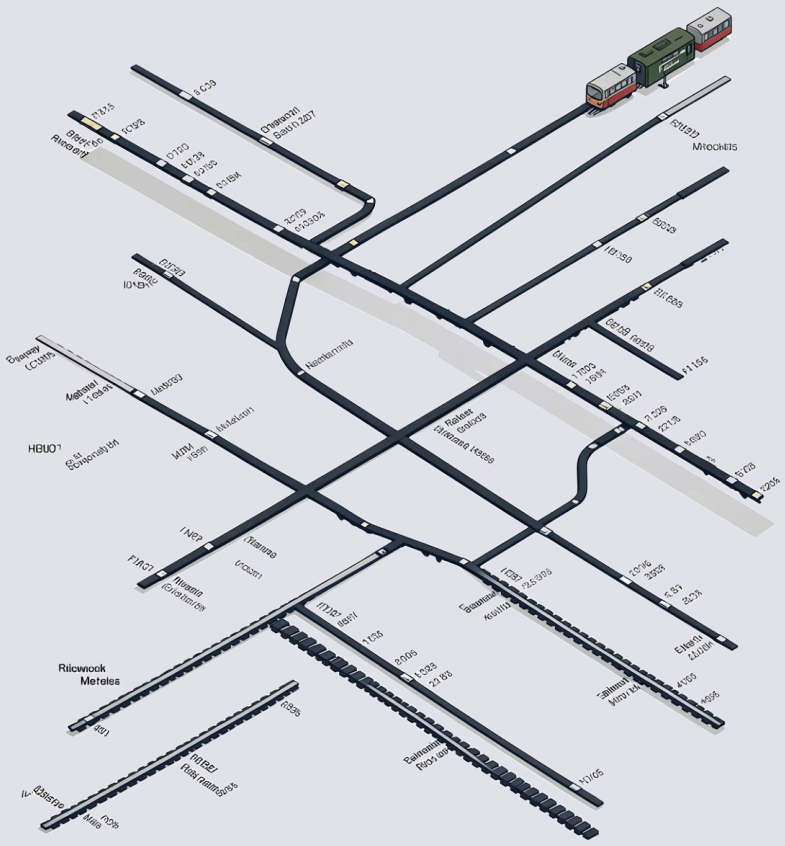
Railway Ticket Reservation & Journey Management System

Database Design & Implementation Project



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Project Scope & Key Entities

Core Entities

Trains, Stations, Routes, Coaches, Seats, Passengers, Bookings, Journeys

Complex Features

Segment-wise seat allocation, waitlists, cancellations, fare management

Business Rules

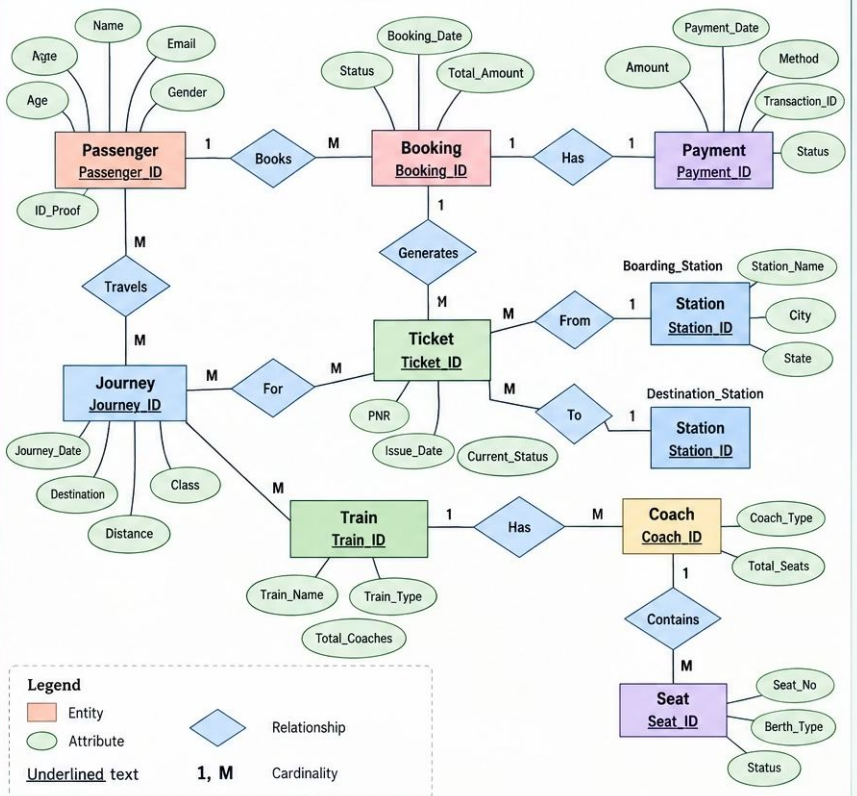
No overlapping seat segments, unique waitlist positions, valid stop sequences

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Railway Ticket Reservation & Journey Management

ER Diagram and Relational Schema

ER Diagram



Relational Schema (Normalized to 3NF)

PASSENGER

PK	Passenger_ID
	Name
	Email
	Phone
	Age
	Gender
	ID_Proof

TRAIN

PK	Train_ID
	Train_Name
	Train_Type
	Total_Coaches

STATION

PK	Station_ID
	Station_Name
	City
	State

JOURNEY

PK	Journey_ID
FK	Train_ID
FK	Source_Station_ID
FK	Destination_Station_ID
	Journey_Date
	Class
	Distance

COACH

PK	Coach_ID
FK	Train_ID
	Coach_Type
	Total_Seats

SEAT

PK	Seat_ID
FK	Coach_ID
	Seat_No
	Berth_Type
	Status

BOOKING

PK	Booking_ID
FK	Passenger_ID
	Booking_Date
	Status
	Total_Amount

TICKET

PK	Ticket_ID
FK	Booking_ID
FK	Journey_ID
	PNR
	Issue_Date
	Current_Status

PAYMENT

PK	Payment_ID
FK	Booking_ID
	Payment_Date
	Amount
	Method
	Transaction_ID
	Status

TICKET_BOARDING

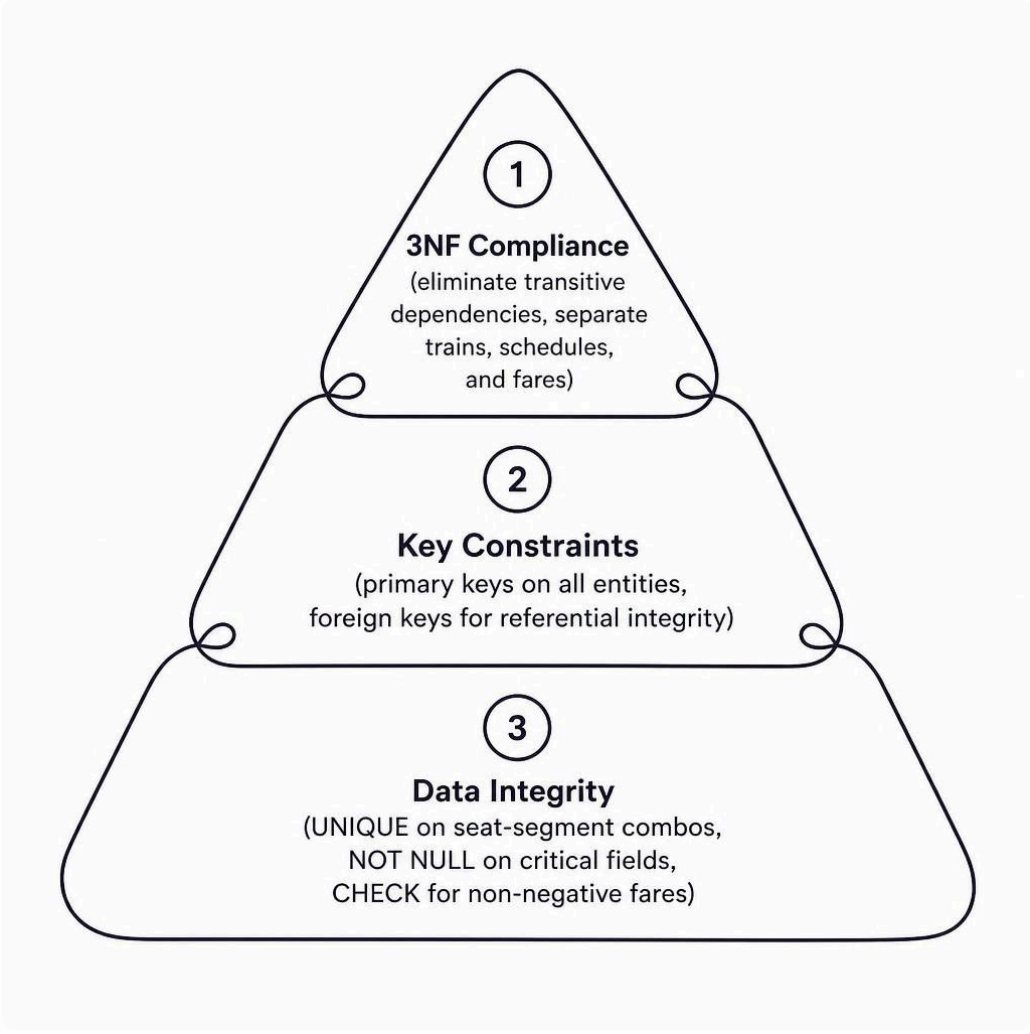
PK, FK	Ticket_ID
PK, FK	Boarding_Station_ID

TICKET_DESTINATION

PK, FK	Ticket_ID
PK, FK	Destination_Station_ID

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Relational Schema & Normalization



3NF Compliance

Eliminate transitive dependencies by separating concerns — trains, schedules, and fares each occupy their own normalized relation.

Key Constraints

Primary keys on all entities; foreign keys enforce referential integrity across every relationship.

Data Integrity

UNIQUE constraints on seat-segment combinations; **NOT NULL** on critical fields; **CHECK** constraints ensure non-negative fares throughout.

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Implementation & SQL Operations



Database Setup

MySQL/PostgreSQL with indexed attributes — `train_id`, `route_id`, `booking_date` — for high-performance search queries.



Core Queries

Train search, segment availability checks, seat conflict detection, waitlist promotion, and revenue reporting.



Transactions

Atomicity guaranteed for booking + payment + seat allocation operations, preventing partial or inconsistent state.

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Frontend Application & Testing

Application Features

- Train search & availability check
- Reservation & waitlist management
- Cancellation & fare calculation

Technology Stack

Java or Python with robust input validation and structured error handling throughout all user flows.

Testing Objectives

Normalization Verification

Confirm 3NF compliance across all relations

Data Integrity

Validate constraints and referential rules under load

Business Rules

Enforce seat overlap prevention and waitlist ordering



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